

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 8 - Linear Stochastic Bandits

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27 September 2026

Overview

- 1 Why care about Bandit problems?
- 2 Stochastic Linear Bandits
- 3 Self-Normalized Bound and Confidence Ellipsoid
- 4 OFUL: Optimism in Face of Uncertainty for Linear Bandits

Introduction and Motivation for Bandits

Problem

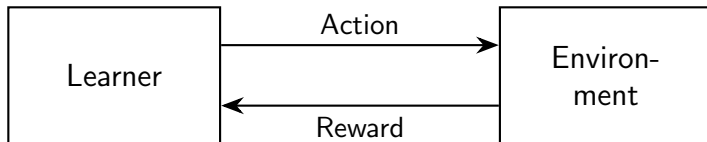
A lot of real-world problems involve making decisions under uncertainty. For instance:

- Deciding which drugs to administer to a patient with comorbidities?
- Which sites to drill for water?
- Which feature change will invite more users and revenue to your app?

Common thread: we only get feedback on the option we *try*, and trying a bad option costs us.

explore vs. exploit

Language of Bandits



One of the oldest works to discuss something adjacent to bandits -
“William R. Thompson. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. Biometrika, 1933”

Language of Bandits

For rounds $t = 1, 2, \dots, n$:

- ① Learner chooses an action A_t from a set $\mathcal{A}$ of available actions. The chosen action is sent to the environment;
- ② The environment generates a response in the form of a real-valued reward $X_t \in \mathbb{R}$, which is sent back to the learner.

The goal of the learner is to maximize the sum of rewards that it receives, $\sum_{t=1}^n X_t$.

At step 2, the agent observes the reward and uses this information to inform which action to take in future rounds. In other words, at time-step t , the learner's decision is based on the trajectory of interactions up to the end of round $t - 1$, i.e., on

$$H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1}).$$

Evaluating the Learner - Regret

Regret

How do we evaluate the learner? Enter **regret**. It is always relative to some action (usually: the best one).

More informally,

Informal definition

Regret relative to action a = Reward accumulated in n rounds when action a is played for all rounds – Reward accumulated by a specific learner / agent in n rounds based on actions chosen by the agent

Small regret $\Rightarrow$ the learner is almost as good as someone who knew the best action from the start.

Finite-armed Stochastic Bandits

Stochastic Bandits

An environment is given by K distributions over the reals, $P_1, \dots, P_K$ and, as discussed before, the learner and the environment interact sequentially. In particular:

For rounds $t = 1, 2, \dots$:

- 1 Based on its past observations (if any), the learner chooses an action $A_t \in [K] \doteq \{1, \dots, K\}$. The chosen action is sent to the environment;
- 2 The environment generates a random reward X_t whose distribution is P_{A_t} (in notation: $X_t \sim P_{A_t}$). The generated reward is sent back to the learner.

Notation: arm i has mean $\mu_i = \mathbb{E}_{X \sim P_i}[X]$; the best mean is $\mu^* = \max_{i \in [K]} \mu_i$.

Objective

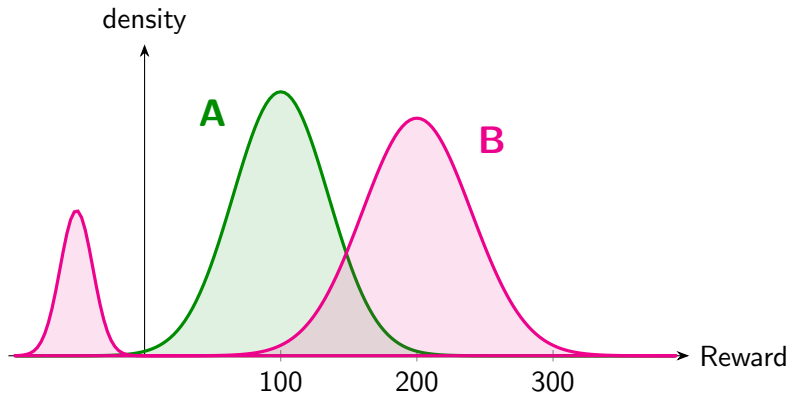
The learner's goal is to maximize its total reward,

$$S_n = \sum_{t=1}^n X_t.$$

Since the X_t are random, S_n is also random and in particular it has a distribution.

Q & A Time

Assume learners A and B with possible total reward distributions shown. Which learner should we prefer?



Regret - Stochastic Bandits case

Answer: B has the larger mean but a risk of losses. The standard (risk-neutral) choice: compare learners through $\mathbb{E}[S_n]$.

Regret

$$R_n = n\mu^* - \mathbb{E}[\sum_{t=1}^n X_t], \quad \mu^* = \max_{i \in [K]} \mu_i$$

The regret is measured relative to the *best* action: playing μ^* in every round.

Goal: sublinear regret, $R_n = o(n)$, i.e. $R_n/n \rightarrow 0$: eventually the learner plays (almost) as well as the best arm.

Why linear structure?

Finite-armed methods learn each arm *separately*: regret grows like $\sqrt{Kn}$. Hopeless when K is huge (millions of articles, continuous doses).

Idea: arms share features

Describe each action by a feature vector $a \in \mathbb{R}^d$ and assume

$$\text{mean reward of } a = \langle a, \theta_* \rangle.$$

Pulling one arm teaches us about *every* arm that shares its features.

Example: news recommendation, a = features of (user, article).

Special case: $\mathcal{A} = \{e_1, \dots, e_K\}$, $\theta_* = (\mu_1, \dots, \mu_K)$ is the K -armed bandit.

Stochastic Linear Bandits

Definition (Linear Stochastic Bandit)

At each round $t = 1, 2, \dots, n$, the learner chooses an action

$$A_t \in \mathcal{A} \subseteq \mathbb{R}^d.$$

There is an unknown parameter $\theta_* \in \mathbb{R}^d$ such that the reward of the chosen action is

$$X_t = \langle A_t, \theta_* \rangle + \varepsilon_t, \quad \langle a, \theta \rangle = a^\top \theta,$$

where ε_t is a random noise term.

After choosing A_t , the learner observes only X_t .

Regret and Optimal Action

The optimal action is

$$A^* \in \arg \max_{a \in \mathcal{A}} \langle a, \theta_* \rangle.$$

The instantaneous regret is $r_t = \langle A^*, \theta_* \rangle - \langle A_t, \theta_* \rangle \geq 0$, and the cumulative regret after n rounds is

$$R_n = \sum_{t=1}^n r_t = \sum_{t=1}^n (\langle A^*, \theta_* \rangle - \langle A_t, \theta_* \rangle).$$

R_n is random (A_t depends on past noise): we bound it *with high probability*.

Recall: optimism and UCB (K arms)

Optimism in the face of uncertainty

Act as if the world is as good as is still *plausible* given the data.

For 1-subgaussian rewards, arm i played $T_i(t-1)$ times:

$$\text{UCB}_i(t) = \hat{\mu}_i(t-1) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}}, \quad A_t = \arg \max_i \text{UCB}_i(t).$$

Why it works: if an arm looks best, either

- it truly is good $\Rightarrow$ little regret, or
- its interval is wide $\Rightarrow$ playing it shrinks the interval (we learn).

Regret: $O(\sqrt{Kn \log n})$.

Towards Confidence Ellipsoids

To use optimism in a linear bandit we need three ingredients:

- ① an estimator $\hat{\theta}_t$ of θ_* (ridge regression)
- ② a confidence set $\mathcal{C}_t$ with $\theta_* \in \mathcal{C}_t$ for all t , (ellipsoid)
- ③ an optimistic choice of A_t using $\mathcal{C}_{t-1}$ (OFUL)

The hard part is (2): the actions A_t depend on past noise, so i.i.d. concentration does not apply.

We will now see the tools that handle adaptively collected data.

History of Interaction

The history before round t is

$$H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1}).$$

Key idea: the learner's decision at time t is allowed to depend on everything observed up to time $t - 1$.

So the central mathematical question becomes:

What does “everything known before time t ” mean?

A σ -algebra is a mathematical information set

Let

$$\mathcal{F}_{t-1} = \sigma(A_1, X_1, \dots, A_{t-1}, X_{t-1}), \quad \mathcal{F}_0 = \{\emptyset, \Omega\}.$$

A σ -algebra tells us which events we can decide (and assign a probability to). Think of $\mathcal{F}_{t-1}$ as:

all questions about the past whose answers are known at time $t - 1$.

For example, for $t \geq 4$,

$$\{X_3 > 1\}, \quad \left\{ \sum_{s=1}^{t-1} X_s > 10 \right\}, \quad \{A_2 = a\} \quad \in \mathcal{F}_{t-1}.$$

Interpretation:

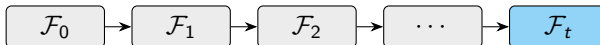
$\mathcal{F}_{t-1} \approx$ “what the learner knows before the new noise arrives.”

A filtration describes growing information

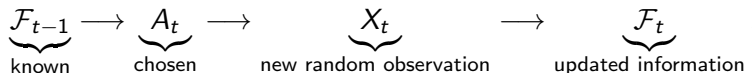
As time passes, information accumulates:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

This sequence is called a **filtration**. A useful picture is



At each round:



The filtration is the mathematical version of the learner's memory.

Conditional expectation = prediction given information

For a random variable Y , $\mathbb{E}[Y \mid \mathcal{F}_{t-1}]$ means:

the best prediction of Y using only information in $\mathcal{F}_{t-1}$.

(best = smallest mean-squared error)

For example, in a linear stochastic bandit, $X_t = A_t^\top \theta_* + \varepsilon_t$. Given $\mathcal{F}_{t-1}$, the action A_t is already determined. Hence

$$\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = A_t^\top \theta_* + \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}].$$

If $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$, then

$$\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = A_t^\top \theta_*.$$

Interpretation: after accounting for everything known from the past, the remaining noise has no predictable direction.

Two rules we will use repeatedly

1. Take out what is known

If Z is $\mathcal{F}_{t-1}$ -measurable, then $\mathbb{E}[ZY \mid \mathcal{F}_{t-1}] = Z \mathbb{E}[Y \mid \mathcal{F}_{t-1}]$.

2. Tower property

$\mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_{t-1}]] = \mathbb{E}[Y]$, and more generally
 $\mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_t] \mid \mathcal{F}_{t-1}] = \mathbb{E}[Y \mid \mathcal{F}_{t-1}]$.

Rule 1 is what we just used ($A_t^\top \theta_*$ is known given $\mathcal{F}_{t-1}$).

Rule 2 lets us peel off one round at a time.

Adaptivity: why conditioning is essential

The learner is **adaptive**: A_t = a function of the past.

Mathematically,

$$A_t \text{ is } \mathcal{F}_{t-1}\text{-measurable.}$$

(True for deterministic learners such as UCB and OFUL; for randomized learners, add A_t to $\mathcal{F}_{t-1}$.)

So, once we condition on $\mathcal{F}_{t-1}$, A_t behaves like a known quantity.
Thus the flow can be summarized as

$$\boxed{\mathcal{F}_{t-1} \longrightarrow A_t \longrightarrow \varepsilon_t \longrightarrow X_t \longrightarrow \mathcal{F}_t.}$$

This is the reason ordinary “IID averaging” intuition is not enough: the actions can depend on previous random rewards.

E.g. stop playing an arm after an unlucky low reward $\Rightarrow$ its sample mean stays biased low.

From zero-mean noise to subgaussian noise

A basic assumption is $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$. This says only:

“the next noise is not predictable from the past.”

For concentration, we need more. We assume conditional R -subgaussian noise:

$$\mathbb{E}[e^{\gamma \varepsilon_t} \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\gamma^2 R^2}{2}\right), \quad \forall \gamma \in \mathbb{R}.$$

The noise may fluctuate, but its MGF is dominated by that of $\mathcal{N}(0, R^2)$: no unusually heavy tails.

Even after seeing the entire past, the new noise is still well behaved.

Remark: letting $\gamma \rightarrow 0$ on both sides forces $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$.

Subgaussian implies zero mean.

Hoeffding's Lemma and Sug-gaussian property

For any random variable X , where $X \in [a, b]$

Setting $a = -R$ and $b = R$, we get $(b - a)^2 = (2R)^2 = 4R^2$, so:

$$\mathbb{E} \left[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1} \right] \leq e^{\frac{\lambda^2 (4B^2)}{8}} = e^{\frac{\lambda^2 B^2}{2}}$$

More intuition on sub-gaussian

Think about a Gaussian random variable $\varepsilon \sim \mathcal{N}(0, R^2)$, for which $\mathbb{E}[e^{\gamma\varepsilon}] = e^{\gamma^2 R^2/2}$ exactly. Its tails decay like e^{-u^2} :

$$P(|\varepsilon| > u) \leq 2 e^{-u^2/(2R^2)}.$$

So very large noise events are extremely unlikely. A random variable is **subgaussian** if its tails are, roughly speaking, **no worse than Gaussian tails**.

Examples

- $\mathcal{N}(0, \sigma^2)$ is σ -subgaussian.
- Zero-mean $\varepsilon \in [a, b]$ is $\frac{b-a}{2}$ -subgaussian (Hoeffding's lemma).
- Not subgaussian: heavy tails (Student- t , Pareto).

From the MGF to a tail bound (Chernoff)

Why the MGF? Exponentiate, then apply Markov bound. For any $\gamma > 0$:

$$P(\varepsilon \geq u) = P(e^{\gamma \varepsilon} \geq e^{\gamma u}) \leq e^{-\gamma u} \mathbb{E}[e^{\gamma \varepsilon}] \leq \exp\left(-\gamma u + \frac{\gamma^2 R^2}{2}\right).$$

This holds for every $\gamma > 0$, so pick the best one, $\gamma = u/R^2$:

$$P(\varepsilon \geq u) \leq \exp\left(-\frac{u^2}{2R^2}\right).$$

Same for $-\varepsilon \Rightarrow$ with probability $\geq 1 - \delta$: $|\varepsilon| \leq R\sqrt{2\log(2/\delta)}$.

MGF bound for all $\gamma =$ tail bound at every scale u .
Note: the best γ depends on the scale. This will matter later.

Martingales

A martingale is a stochastic process in which the expected value of the next observation, given all prior observations, is equal to the most recent value. Define $M_0 = 0$ and $M_t = \sum_{s=1}^t \varepsilon_s$.

If $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$ (and ε_t is $\mathcal{F}_t$ -measurable, integrable), then

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1} + \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = M_{t-1}.$$

Hence $(M_t, \mathcal{F}_t)_{t \geq 0}$ is a martingale.

Interpretation:

given everything we know now, the expected next value is the current value.

Subgaussianity strengthens this as zero mean only gives variance based tails, which decays polynomially. But sub-gaussian property gives exponential tails

martingale diff. + subgaussian increments $\implies$ strong concentration of M_t .

Supermartingales: the recipe

$(D_t, \mathcal{F}_t)$ is a **supermartingale** if

$$\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] \leq D_{t-1} \quad (\text{expected to go down}).$$

Tower rule, round by round:

$$\mathbb{E}[D_t] = \mathbb{E}[\mathbb{E}[D_t \mid \mathcal{F}_{t-1}]] \leq \mathbb{E}[D_{t-1}] \leq \cdots \leq \mathbb{E}[D_0].$$

Recipe for concentration

Build a nonnegative supermartingale with $D_0 = 1$ that is *large* when the noise is large. Then $\mathbb{E}[D_t] \leq 1$, and by Markov

$$P(D_t \geq 1/\delta) \leq \delta.$$

Everything that follows applies this recipe.

Sums with predictable weights

Now consider a weighted sum

$$S_t = \sum_{s=1}^t a_s \varepsilon_s,$$

where a_s is $\mathcal{F}_{s-1}$ -measurable (*predictable*: fixed before ε_s arrives).

Given $\mathcal{F}_{s-1}$, a_s is a constant. Subgaussianity with γa_s in place of γ :

$$\mathbb{E}\left[e^{\gamma a_s \varepsilon_s} \mid \mathcal{F}_{s-1}\right] \leq e^{\gamma^2 R^2 a_s^2 / 2} \iff \boxed{\mathbb{E}\left[e^{\gamma a_s \varepsilon_s - \frac{\gamma^2 R^2}{2} a_s^2} \mid \mathcal{F}_{s-1}\right] \leq 1.}$$

Each round's “exponential increment” has conditional mean at most 1.

The exponential supermartingale

Multiply the increments:

$$D_t = \exp \left(\gamma S_t - \frac{\gamma^2 R^2}{2} \sum_{s=1}^t a_s^2 \right), \quad D_0 = 1.$$

One step: $D_t = D_{t-1} e^{\gamma a_t \varepsilon_t - \frac{\gamma^2 R^2}{2} a_t^2}$, so by rule 1

$$\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = D_{t-1} \mathbb{E} \left[e^{\gamma a_t \varepsilon_t - \frac{\gamma^2 R^2}{2} a_t^2} \mid \mathcal{F}_{t-1} \right] \leq D_{t-1}.$$

D_t is a nonnegative supermartingale, so

$$\mathbb{E} \left[\exp \left(\gamma S_t - \frac{\gamma^2 R^2}{2} \sum_{s=1}^t a_s^2 \right) \right] \leq 1 \quad \forall \gamma \in \mathbb{R}.$$

The random scale $\sum_s a_s^2$ stays *inside* the expectation.

The catch: the scale is random

If the weights a_s are **deterministic**, $\sum_s a_s^2$ comes out:

$$\mathbb{E}[e^{\gamma S_t}] \leq \exp\left(\frac{\gamma^2 R^2}{2} \sum_{s=1}^t a_s^2\right) \Rightarrow S_t \text{ is } R\sqrt{\sum_s a_s^2}\text{-subgaussian.}$$

In a bandit, a_s depends on past noise, so $\sum_s a_s^2$ is **random**:

- the display above is no longer valid (its right side is random);
- Chernoff needs $\gamma = u/(R^2 \sum_s a_s^2)$, which depends on the data. We cannot pick γ after looking.

Fix: measure the noise in units of its own random scale:
self-normalized bounds.

Ridge regression

For $X_t = A_t^\top \theta_* + \varepsilon_t$, estimate θ_* by

$$\hat{\theta}_t = \arg \min_{\theta \in \mathbb{R}^d} \sum_{s=1}^t (X_s - A_s^\top \theta)^2 + \lambda \|\theta\|_2^2.$$

Setting the gradient to zero:

$$\left(\lambda I + \sum_{s=1}^t A_s A_s^\top \right) \theta = \sum_{s=1}^t A_s X_s.$$

$$\hat{\theta}_t = V_t^{-1} \sum_{s=1}^t A_s X_s, \quad V_t = \lambda I + G_t, \quad G_t = \sum_{s=1}^t A_s A_s^\top.$$

V_t : regularized Gram (“information”) matrix. $\lambda > 0$ makes it invertible from the start: $V_0 = \lambda I$, $\hat{\theta}_0 = 0$.

The linear-bandit noise term

Substituting the model into $\sum_s A_s X_s$, with $Z_t := \sum_{s=1}^t A_s \varepsilon_s$:

$$\sum_{s=1}^t A_s X_s = \sum_{s=1}^t A_s A_s^\top \theta_* + \sum_{s=1}^t A_s \varepsilon_s = (V_t - \lambda I) \theta_* + Z_t.$$

Multiplying by V_t^{-1} :

$$\hat{\theta}_t - \theta_* = \underbrace{-\lambda V_t^{-1} \theta_*}_{\text{bias (regularization)}} + \underbrace{V_t^{-1} Z_t}_{\text{noise}}.$$

The random part is

$$V_t^{-1} Z_t = V_t^{-1} \sum_{s=1}^t A_s \varepsilon_s.$$

Linear-bandit noise: one direction at a time

Projecting onto any *fixed* direction u :

$$u^\top Z_t = \sum_{s=1}^t (u^\top A_s) \varepsilon_s.$$

Since A_s is chosen using the past, $u^\top A_s$ is $\mathcal{F}_{s-1}$ -measurable.

Therefore the subgaussian argument applies with $a_s = u^\top A_s$.

Since $\sum_s (u^\top A_s)^2 = u^\top G_t u$:

$$\mathbb{E} \left[\exp \left(\gamma u^\top Z_t - \frac{\gamma^2 R^2}{2} u^\top G_t u \right) \right] \leq 1.$$

Works for every fixed u . Not enough, though (next slide).

Why one direction is not enough

Problem 1: we need control in *all* directions at once (every action $a \in \mathcal{A}$ is a direction), and there are infinitely many.

Problem 2: the scale $u^\top G_t u$ in direction u is random.

Both point to the same target. By Cauchy–Schwarz,

$$\sup_{u \neq 0} \frac{u^\top Z_t}{\|u\|_{V_t}} = \|Z_t\|_{V_t^{-1}}, \quad \|z\|_M := \sqrt{z^\top M z}.$$

Goal: bound the self-normalized noise $\|Z_t\|_{V_t^{-1}}$:
the worst direction, measured in units of how much we explored it.

One supermartingale for every vector x

Merge γ and u into one vector $x = \gamma Ru \in \mathbb{R}^d$:

$$D_t(x) = \exp \left(\frac{x^\top Z_t}{R} - \frac{1}{2} \|x\|_{G_t}^2 \right).$$

Each round's factor has conditional mean ≤ 1 (subgaussian with $\gamma = x^\top A_s / R$), so

$$\mathbb{E}[D_t(x)] \leq 1 \quad \text{for every fixed } x.$$

If we could plug in the best $x = G_t^{-1} Z_t / R$ (G_t invertible):

$$\sup_x \log D_t(x) = \frac{\|Z_t\|_{G_t^{-1}}^2}{2R^2} \quad (\text{our target!})$$

But the best x depends on the data: choosing x after looking breaks $\mathbb{E} \leq 1$.

Method of mixtures: average instead of maximize

We cannot maximize over x , but we can **average** over x with fixed weights. Let φ be the density of $\mathcal{N}(0, \lambda^{-1}I)$ and

$$\bar{D}_t = \int_{\mathbb{R}^d} D_t(x) \varphi(x) dx.$$

φ does not depend on the data, so applying fubini (double integral can be computed in iterative fasion)

$$\mathbb{E}[\bar{D}_t \mid \mathcal{F}_{t-1}] = \int \mathbb{E}[D_t(x) \mid \mathcal{F}_{t-1}] \varphi(x) dx \leq \bar{D}_{t-1}.$$

$\bar{D}_t$ is again a nonnegative supermartingale, with $\bar{D}_0 = 1$.

Intuition (Laplace): an average of $e^{f(x)}$ is dominated by the region near the maximizer.

Computing the mixture

With $b = Z_t/R$ and $V_t = G_t + \lambda I$, complete the square:

$$x^\top b - \frac{1}{2} x^\top G_t x - \frac{\lambda}{2} \|x\|^2 = -\frac{1}{2} (x - V_t^{-1} b)^\top V_t (x - V_t^{-1} b) + \frac{1}{2} \|b\|_{V_t^{-1}}^2.$$

Use $\int e^{-\frac{1}{2}(x-m)^\top V(x-m)} dx = (2\pi)^{d/2} \det(V)^{-1/2}$ and the constant $(\lambda/2\pi)^{d/2}$ of φ :

$$\bar{D}_t = \left(\frac{\lambda^d}{\det V_t} \right)^{1/2} \exp \left(-\frac{\|Z_t\|_{V_t^{-1}}^2}{2R^2} \right).$$

The ridge parameter λ reappears: the mixing precision turns G_t into V_t .

The self-normalized bound

Markov bound: $P(\bar{D}_t \geq 1/\delta) \leq \delta \mathbb{E}[\bar{D}_t] \leq \delta$.

Ville's maximal inequality (nonnegative supermartingale, $\bar{D}_0 = 1$):

$$P(\exists t \geq 0 : \bar{D}_t \geq 1/\delta) \leq \delta.$$

Taking logs of $\bar{D}_t < 1/\delta$:

Theorem (Abbasi-Yadkori, Pál, Szepesvári, 2011)

With probability at least $1 - \delta$, simultaneously for all $t \geq 0$,

$$\|Z_t\|_{V_t^{-1}} \leq R \sqrt{2 \log(1/\delta) + \log(\det V_t / \lambda^d)}.$$

Anytime: holds for all rounds at once, as OFUL uses it in every round.

From concentration to a confidence ellipsoid

Recall $\hat{\theta}_t - \theta_* = -\lambda V_t^{-1} \theta_* + V_t^{-1} Z_t$. With $\|V_t^{-1} z\|_{V_t} = \|z\|_{V_t^{-1}}$:

$$\|\hat{\theta}_t - \theta_*\|_{V_t} \leq \lambda \|\theta_*\|_{V_t^{-1}} + \|Z_t\|_{V_t^{-1}} \leq \sqrt{\lambda} S + \|Z_t\|_{V_t^{-1}},$$

using $V_t \succeq \lambda I$ and $\|\theta_*\|_2 \leq S$. With the self-normalized bound, with probability $\geq 1 - \delta$, for all $t \geq 0$:

$$\|\hat{\theta}_t - \theta_*\|_{V_t} \leq \beta_t(\delta) := R \sqrt{2 \log \frac{1}{\delta} + \log \frac{\det V_t}{\lambda^d}} + \sqrt{\lambda} S.$$

Confidence ellipsoid:

$$\mathcal{C}_t = \{\theta \in \mathbb{R}^d : \|\theta - \hat{\theta}_t\|_{V_t} \leq \beta_t(\delta)\}.$$

Uncertainty about a single action

For a candidate action a , Cauchy–Schwarz in the V_t -geometry:

$$|a^\top(\hat{\theta}_t - \theta_*)| = |(V_t^{-1/2}a)^\top V_t^{1/2}(\hat{\theta}_t - \theta_*)| \leq \|a\|_{V_t^{-1}} \|\hat{\theta}_t - \theta_*\|_{V_t}.$$

Consequently (on the good event),

$$|a^\top(\hat{\theta}_t - \theta_*)| \leq \beta_t \|a\|_{V_t^{-1}}.$$

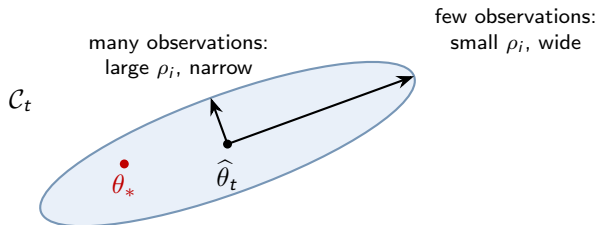
$$\text{uncertainty about } a^\top \theta_* \approx \beta_t \|a\|_{V_t^{-1}}.$$

Sanity check (K arms): $a = e_i$, $V_t = \lambda I + \text{diag}(T_i(t))$

$\Rightarrow \|e_i\|_{V_t^{-1}} = 1/\sqrt{\lambda + T_i(t)}$: the familiar $1/\sqrt{\#\text{pulls}}$ width of UCB.

Geometry of the confidence ellipsoid

Eigen-decompose $V_t = \sum_i \rho_i v_i v_i^\top$. Along v_i , $\mathcal{C}_t$ has half-width $\beta_t / \sqrt{\rho_i}$.



$\log(\det V_t / \lambda^d) = \sum_i \log(\rho_i / \lambda)$ grows only like $d \log t$, so $\beta_t \approx \sqrt{d \log t}$.

The matrix V_t^{-1} records where the learner has and has not collected information.

Optimism in face of Uncertainty

Among all plausible environments that are consistent with the data observed so far, the learner should be optimistic and act as if the environment is the best possible one.

Optimism in the face of uncertainty (linear)

$\mathcal{C}_{t-1}$ contains every parameter still plausible after $t - 1$ rounds. Be optimistic about the action *and* the parameter:

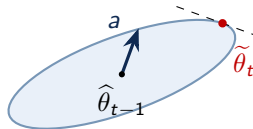
$$(A_t, \tilde{\theta}_t) = \arg \max_{a \in \mathcal{A}, \theta \in \mathcal{C}_{t-1}} \langle a, \theta \rangle$$

Optimistic value of a :

$$\text{UCB}_t(a) = \max_{\theta \in \mathcal{C}_{t-1}} \langle a, \theta \rangle,$$

the best reward a could plausibly have.

Round t only uses data from rounds $1, \dots, t - 1$.



$\tilde{\theta}_t$: point of $\mathcal{C}_{t-1}$
furthest in direction a

Solving the inner maximization

Fix a . Every $\theta \in \mathcal{C}_{t-1}$ can be written as

$$\theta = \hat{\theta}_{t-1} + V_{t-1}^{-1/2} w, \quad \|w\|_2 \leq \beta_{t-1}.$$

Then, by Cauchy–Schwarz,

$$\langle a, \theta \rangle = \langle a, \hat{\theta}_{t-1} \rangle + \langle V_{t-1}^{-1/2} a, w \rangle \leq \langle a, \hat{\theta}_{t-1} \rangle + \beta_{t-1} \|a\|_{V_{t-1}^{-1}},$$

with equality at $w \propto V_{t-1}^{-1/2} a$. Hence

$$\boxed{\text{UCB}_t(a) = \underbrace{\langle a, \hat{\theta}_{t-1} \rangle}_{\text{exploit}} + \underbrace{\beta_{t-1} \|a\|_{V_{t-1}^{-1}}}_{\text{exploration bonus}}}$$

Same shape as UCB: estimate + radius $\times$ width in direction a .

OFUL: the algorithm

OFUL (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Input: $\lambda > 0$, $\delta \in (0, 1)$, R , S . **Init:** $V_0 = \lambda I$, $b_0 = 0$, $\hat{\theta}_0 = 0$.

for $t = 1, 2, \dots, n$ **do**

- ① $\beta_{t-1} = R\sqrt{2 \log(1/\delta) + \log(\det V_{t-1}/\lambda^d)} + \sqrt{\lambda} S$

- ② $A_t = \arg \max_{a \in \mathcal{A}} \langle a, \hat{\theta}_{t-1} \rangle + \beta_{t-1} \|a\|_{V_{t-1}^{-1}}$

- ③ play A_t , observe X_t

- ④ $V_t = V_{t-1} + A_t A_t^\top$, $b_t = b_{t-1} + A_t X_t$, $\hat{\theta}_t = V_t^{-1} b_t$

Exploration is driven by uncertainty. The width is large exactly in the directions we have not explored.

Implementation notes

Rank-one updates, $O(d^2)$ per round:

- Sherman–Morrison: $V_t^{-1} = V_{t-1}^{-1} - \frac{V_{t-1}^{-1} A_t A_t^\top V_{t-1}^{-1}}{1 + \|A_t\|_{V_{t-1}^{-1}}^2}$
- Matrix determinant lemma: $\det V_t = \det V_{t-1} (1 + \|A_t\|_{V_{t-1}^{-1}}^2)$
(we reuse this in the analysis)

Action selection:

- finite $\mathcal{A}$: $O(|\mathcal{A}| d^2)$ per round;
- convex $\mathcal{A}$ (e.g. a polytope): the index is *convex* in a , so maximizing it over $\mathcal{A}$ is a non-convex problem and can be computationally hard (Dani et al., 2008).

In practice: LinUCB (Li et al., 2010) uses the same index with a tuned constant α in place of β_{t-1} .

Assumptions for the regret bound

- bounded actions: $\|a\|_2 \leq L$ for all $a \in \mathcal{A}$
- bounded parameter: $\|\theta_*\|_2 \leq S$ (S known)
- bounded mean rewards: $|\langle a, \theta_* \rangle| \leq 1 \Rightarrow r_t \leq 2$
- ε_t conditionally R -subgaussian; A_t is $\mathcal{F}_{t-1}$ -measurable

Good event: $\mathcal{E} = \{\theta_* \in \mathcal{C}_t \text{ for all } t \geq 0\}$, $P(\mathcal{E}) \geq 1 - \delta$.

We prove the regret bound *deterministically* on $\mathcal{E}$.

Shorthand: $w_t = \|A_t\|_{V_{t-1}^{-1}}$, the width of the played action.

Regret of OFUL

Theorem

With probability at least $1 - \delta$,

$$R_n \leq \sqrt{8 n d \beta_n^2 \log \left(1 + \frac{n L^2}{\lambda d} \right)},$$

where (w.l.o.g. $\beta_n \geq 1$)

$$\beta_n \leq R \sqrt{2 \log(1/\delta) + d \log \left(1 + \frac{n L^2}{\lambda d} \right)} + \sqrt{\lambda} S.$$

In words: $\beta_n = O(\sqrt{d \log n})$, so $R_n = O(d \sqrt{n} \log n) = \tilde{O}(d \sqrt{n})$.

Step 1: optimism bounds regret by the width

On $\mathcal{E}$, (A^*, θ_*) is a feasible pair in the OFUL maximization, so $\langle A_t, \tilde{\theta}_t \rangle \geq \langle A^*, \theta_* \rangle$. Therefore

$$\begin{aligned}
 r_t &= \langle A^*, \theta_* \rangle - \langle A_t, \theta_* \rangle \leq \langle A_t, \tilde{\theta}_t - \theta_* \rangle \\
 &\leq \|A_t\|_{V_{t-1}^{-1}} \|\tilde{\theta}_t - \theta_*\|_{V_{t-1}} \leq 2\beta_{t-1} w_t,
 \end{aligned}$$

since $\tilde{\theta}_t, \theta_* \in \mathcal{C}_{t-1}$ are both within β_{t-1} of $\hat{\theta}_{t-1}$ (triangle inequality).

regret in round $t \leq 2 \times$ our uncertainty about the action we played

Either the optimistic action is good, or it is uncertain, and then we learn about it.

Step 2: every play shrinks the ellipsoid

Playing A_t adds $A_t A_t^\top$ to the information matrix:

$$V_t = V_{t-1} + A_t A_t^\top = V_{t-1}^{1/2} (I + v v^\top) V_{t-1}^{1/2}, \quad v = V_{t-1}^{-1/2} A_t.$$

Since $\det(I + v v^\top) = 1 + \|v\|^2 = 1 + w_t^2$:

$$\boxed{\det V_t = \det V_{t-1} (1 + w_t^2)} \Rightarrow \sum_{t=1}^n \log(1 + w_t^2) = \log \frac{\det V_n}{\lambda^d}.$$

For a fixed radius, $\text{vol}(\mathcal{C}_t) \propto (\det V_t)^{-1/2}$: a wide play (w_t large) shrinks the ellipsoid a lot.

Step 3: the elliptical potential lemma

Total shrinkage is limited: $\det V_n$ cannot grow too fast. AM–GM on the eigenvalues:

$$\det V_n \leq \left(\frac{\text{tr } V_n}{d} \right)^d \leq \left(\frac{d\lambda + nL^2}{d} \right)^d,$$

$$\Rightarrow \log \frac{\det V_n}{\lambda^d} \leq d \log \left(1 + \frac{nL^2}{\lambda d} \right).$$

With $\min(1, x) \leq 2 \log(1 + x)$ for $x \geq 0$:

Elliptical potential lemma

$$\sum_{t=1}^n \min(1, w_t^2) \leq 2 \log \frac{\det V_n}{\lambda^d} \leq 2d \log \left(1 + \frac{nL^2}{\lambda d} \right).$$

Widths can be large only $O(d \log n)$ times: there are only d directions to explore.

Step 4: putting it together

$r_t \leq 2$ and $r_t \leq 2\beta_{t-1}w_t$, with $\beta_{t-1} \leq \beta_n$ (β_t is nondecreasing, $\beta_n \geq 1$):

$$r_t \leq \min(2, 2\beta_{t-1}w_t) \leq 2\beta_n \min(1, w_t).$$

Cauchy–Schwarz, then the potential lemma:

$$\begin{aligned}
 R_n = \sum_{t=1}^n r_t &\leq \sqrt{n \sum_{t=1}^n r_t^2} \leq 2\beta_n \sqrt{n \sum_{t=1}^n \min(1, w_t^2)} \\
 &\leq 2\beta_n \sqrt{2nd \log \left(1 + \frac{nL^2}{\lambda d}\right)}.
 \end{aligned}$$

This is the theorem.

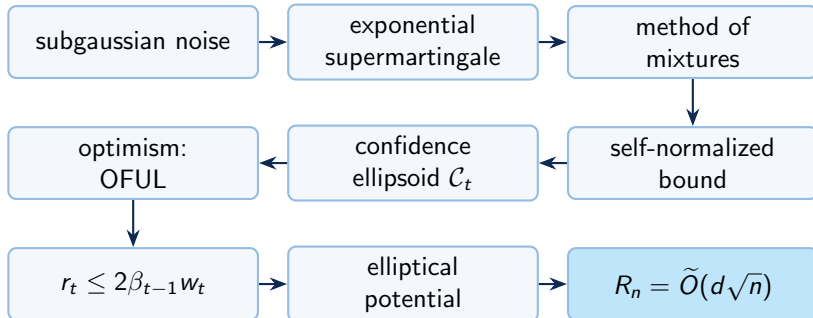
Expected regret: $\delta = 1/n$ gives $\mathbb{E}[R_n] \leq (\text{same bound}) + 2$.

Interpreting the bound

$$R_n = \tilde{O}(d\sqrt{n})$$

- **Sublinear:** average regret $R_n/n \rightarrow 0$.
- **Independent of $|\mathcal{A}|$:** works for infinitely many actions.
- **Two factors of $\sqrt{d}$:**
 - $\beta_n \approx \sqrt{d \log n}$: the ellipsoid must cover d directions;
 - $\sum_t \min(1, w_t^2) \lesssim d \log n$: there are d directions to explore.

Summary



Adaptive data $\Rightarrow$ martingale tools $\Rightarrow$ ellipsoid $\Rightarrow$ optimism explores exactly where V_t^{-1} is large.

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